VF-E27 Spray Test

Test Method for Spray Test	Test method # VF- E27 REF #:
Date Issued: September 15, 2022 Revision # :0	Reason for Revision: First launch

PURPOSE AND SCOPE: For determining the resistance of any fabric, which might or might not have been given a water-resistant or water-repellent finish, to surface wetting by water.

REFERENCE: ISO 4920, AATCC 22

EQUIPMENT AND MATERIAL:

1. **Spray device**, consisting of a funnel of (150 ± 5) mm diameter held vertically, with a metal nozzle connected to the end of the stem by rubber tubing of 10 mm bore (see Figure 1).
2. **Metal spray nozzle**2)**,** having a convex face with 19 holes of (0,86 ± 0,05) mm diameter (see Figure 2). The holes are distributed over the face of the nozzle. The duration of flow for the specified volume of (250 ± 2) ml of water poured into the funnel shall be between 25 s and 30 s.
3. **Specimen holder**, consisting of two wood or metal rings, which fit into each other, one of inside diameter (155 ± 5) mm and another of outside diameter (155 ± 5) mm (for example an embroidery hoop), into which the specimen can be secured (see Figure 1).
The rings should, when in position for a test, rest on a suitable support so that it is inclined at an angle of 45° with the center of the test area (150 ± 2) mm below the center of the face of the spray nozzle.
4. **Water,** distilled or fully deionized, at (20 ± 2) °C or (27 ± 2) °C, or other temperatures, as agreed between interested parties. The temperatures used shall be stated in the test report.
5. **Graduated cylinder**, 250 ml, class B.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Take at least three test specimens, each at least 180 mm square, from different places in the fabric so that they represent the material as fully as possible and can be secured by the specimen holder. Do not take test specimens from places with creases or fold marks.

REPETITIONS: Test three samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

1. Condition the test specimens for at least 4 h.
2. After conditioning, mount the test specimen securely on the specimen holder with the face of the fabric uppermost on the hoop support. Place the holder on the support as specified in equipment. Unless otherwise stated in the material specification, the specimen shall be orientated so that the warp, or length, direction is parallel to the flow of water down the specimen.

3. Pour (250 ± 2) ml of water into the funnel quickly, but steadily so that the spraying shall be continuous once it has commenced. The duration of flow shall be between 25 s and 30 s.
4. Immediately after the spray has ceased, remove the holder with its specimen. Hold the specimen holder by the bottom edge, with the plane of the fabric almost horizontal and the fabric face down. Tap the holder smartly against a solid object with the fabric facing the object. Rotate the specimen holder 180° and tap it smartly once more.
5. Immediately after tapping, with the specimen still on the holder, rate the degree of wetting of the face of the specimen only, according to evaluate.
6. Repeat steps 1 through 5 for all specimens.

Evaluation

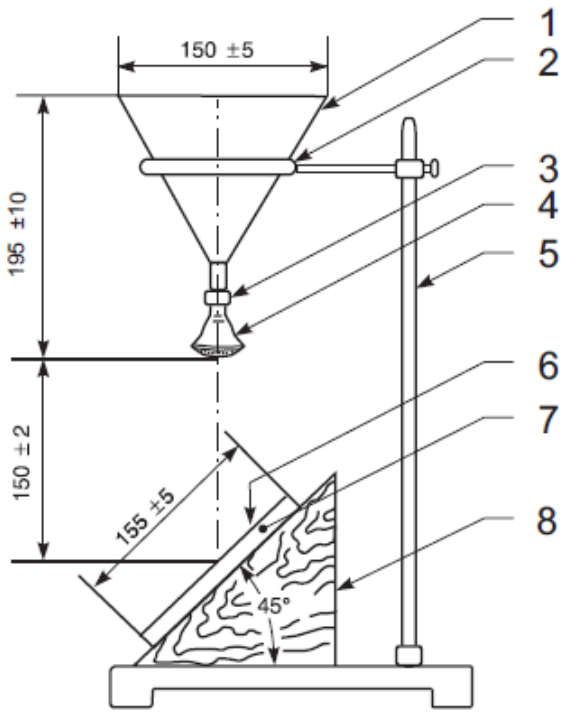
1. Assign a rating to each test specimen which best corresponds to the degree of wetting as described in the spray rating scale or the photographic spray rating standard scale (see Annex A and Figure A.1). Intermediate ratings can be used in increments of 0,5 for a rating corresponding to ISO 1 or AATCC 50, or higher.
Photographic standards are not entirely satisfactory for dark-colored fabrics, for which more reliance should be placed on verbal descriptions.
2. The spray rating scale is defined as follows:
0 — Complete wetting of the entire face of the specimen.
1 — Complete wetting of the entire specimen face beyond the spray points.
2 — Partial wetting of the specimen face beyond the spray points.
3 — Wetting of specimen face at spray points.
4 — Slight random sticking or wetting of the specimen face.
5 — No sticking or wetting of the specimen face.

Test report

The test report shall include the following information:
a) a reference to this International Standard.
b) atmosphere used.
c) temperature of the water.
d) spray rating for each specimen tested.
e) all data needed to identify the sample(s) tested.
f) any deviation from the procedure specified

VF-E27 Spray Test

Dimensions in millimetres unless otherwise indicated



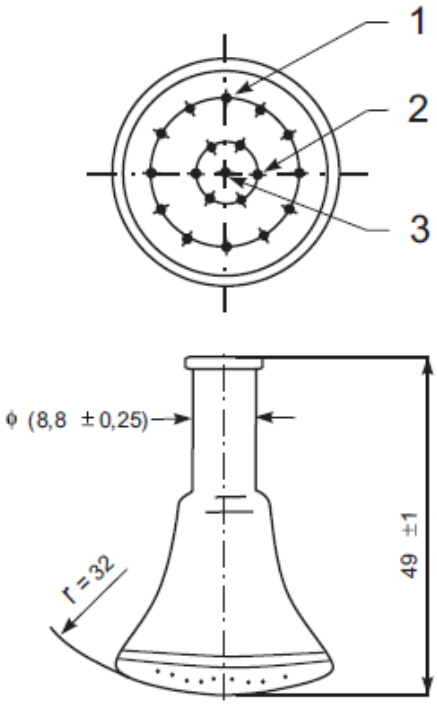
Key

1 funnel
2 ring support
3 rubber tubing
4 spray nozzle
5 stand
6 specimen
7 specimen holder
8 support

Figure 1 — Apparatus for spray test

VF-E27 Spray Test

Dimensions in millimetres



Key

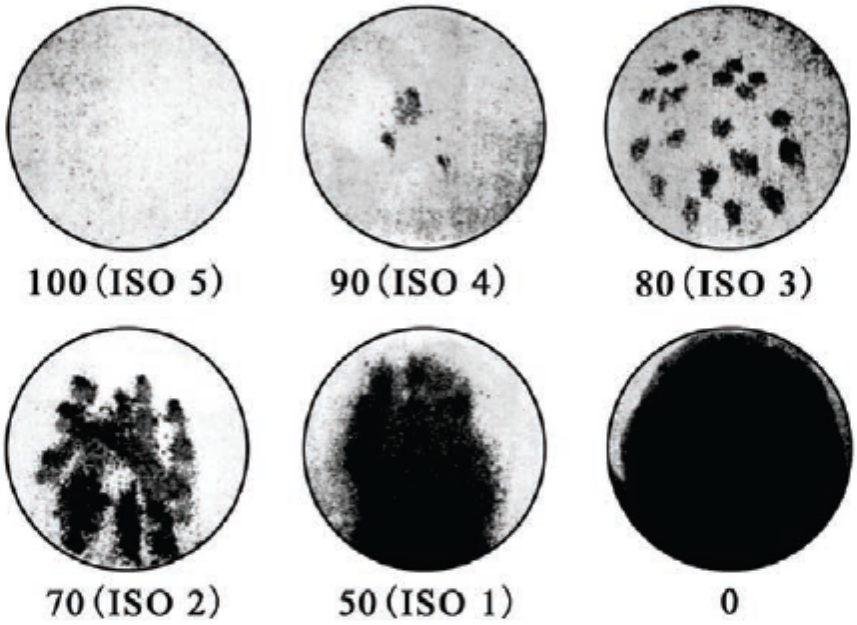
1 12 holes ∅(0,86 ± 0,05) on a circle ∅(21 ± 0,5)
2 6 holes ∅(0,86 ± 0,05) on a circle ∅(10 ± 0,5)
3 1 hole ∅(0,86 ± 0,05) on centre

Figure 2 — Spray nozzle details

Annex A

The ISO descriptive rating scale corresponds to the AATCC photographic scale[3] as follows:

ISO 0 = AATCC 0,
ISO 1 = AATCC 50,
ISO 2 = AATCC 70,
ISO 3 = AATCC 80,
ISO 4 = AATCC 90,
ISO 5 = AATCC 100



100	no sticking or wetting of the specimen
90	slight random sticking or wetting of the specimen face
80	wetting of specimen face at spray points
70	partial wetting of the specimen face beyond the spray points
50	complete wetting of the entire specimen face beyond the spray points
0	complete wetting of the entire face of the specimen

Coloured water used for photographic effect

Figure A.1 — ISO spray test rating chart based on the AATCC photographic scale